Synopsis

- **Project Title :-** EduReport an annual report portal
- **Guide :-** Prof.A.S.Phalle
- **Area / Domain of Project :-** Web development and Machine learning

(1) **Introduction** :- Educational institutes generate a large volume of data annually across various departments—academic, research, finance, infrastructure, and extracurricular activities. Preparing a comprehensive and accurate annual report from this data is often time-consuming, inconsistent, and manual.

(2) **Motivation :-** Preparing annual reports in educational institutes is often a manual, repetitive, and error-prone process. Each department works in silos, leading to delays, data inconsistency, and lack of standardization. There is a growing need for a unified, efficient, and automated solution that simplifies data collection, enhances collaboration, and presents insights in a structured format. This motivated the development of EduReport — a smart portal that streamlines annual report creation and supports informed decision-making through accurate and well-organized data.

(3) **Existing System** :- In most educational institutes, the annual report preparation process is manual and decentralized. Each department typically collects its own data using spreadsheets, word documents, or emails. These reports are then individually submitted to a central authority for compilation. This approach lacks standardization, consumes significant time, and increases the chances of data duplication, inconsistency, and human error.

(4) **Drawbacks/limitations/challenges in Existing System** :-

- Manual Process: Time-consuming and prone to errors and duplication.
- Lack of Standardization: Inconsistent data formats across departments.
- No Collaboration Tools: Limited scope for real-time editing or feedback.
- Poor Data Visualization: Absence of graphs or dashboards for insights.
- Version Control Issues: Difficult to manage and track document updates.
- Security Risks: Unsecured sharing of sensitive institutional data.

(5) **Problem Statement :-**

Develop a portal to streamline data collection and generate accurate, consistent annual reports for educational institutions.

(6) Feasibility Study :-

The feasibility study evaluates the technical, operational, and economic aspects of EduReport. Technically, the project is feasible as the required technologies (React, Python/Django/Flask, MongoDB, ML frameworks) are readily available. Operational feasibility is ensured since institutes already collect such data and users are familiar with web portals. Economically, the system reduces manual workload and long-term costs, making it cost-effective.

(7) Project Scope :-

1. Data collection and integration
2. Data analysis and visualization
3. Report generation
4. Collaboration and feedback
5. User authentication and role management

(8) Requirement Engineering :-

Requirement engineering involves the systematic process of eliciting, analyzing, documenting, and validating requirements. For EduReport, this includes functional requirements (user management, activity submission, ML verification, reporting) and non-functional requirements (usability, performance, security, scalability). Stakeholder discussions and iterative feedback will guide the refinement of requirements.

(9) Software Requirement Specification (SRS) :-

- **Functional Requirements**

1. User Authentication & Roles – Secure login, role-based access.
2. Data Collection & Integration – Import from databases/spreadsheets, system integration, manual entry.
3. Analysis & Visualization – KPIs dashboards, charts, graphs.
4. Report Generation – Automated PDF/HTML reports, customizable templates, multimedia support.
5. Collaboration & Feedback – Editing, reviews, feedback, version control.
6. User Experience – Intuitive UI, responsive design, multilingual support.

- **Non-Functional Requirements**

1. Performance – Fast, responsive, handles large data.
2. Security – Compliance (GDPR/FERPA), strong protection.
3. Scalability – Works for different institutes, supports growth.

- **System Architecture**

1. Frontend – HTML, CSS, JS (Bootstrap/Materialize).
2. Backend – Python/Java with MySQL/MongoDB.
3. Integration – APIs, import/export (CSV, Excel).

- **Data Management**

1. Storage – Structured DB, backup & recovery.
2. Security – Encryption, access control, audits.

- **Testing**

1. Unit Testing – Individual modules.
2. Integration Testing – External systems/data.
3. User Acceptance – End-user validation.

- **Deployment & Maintenance**

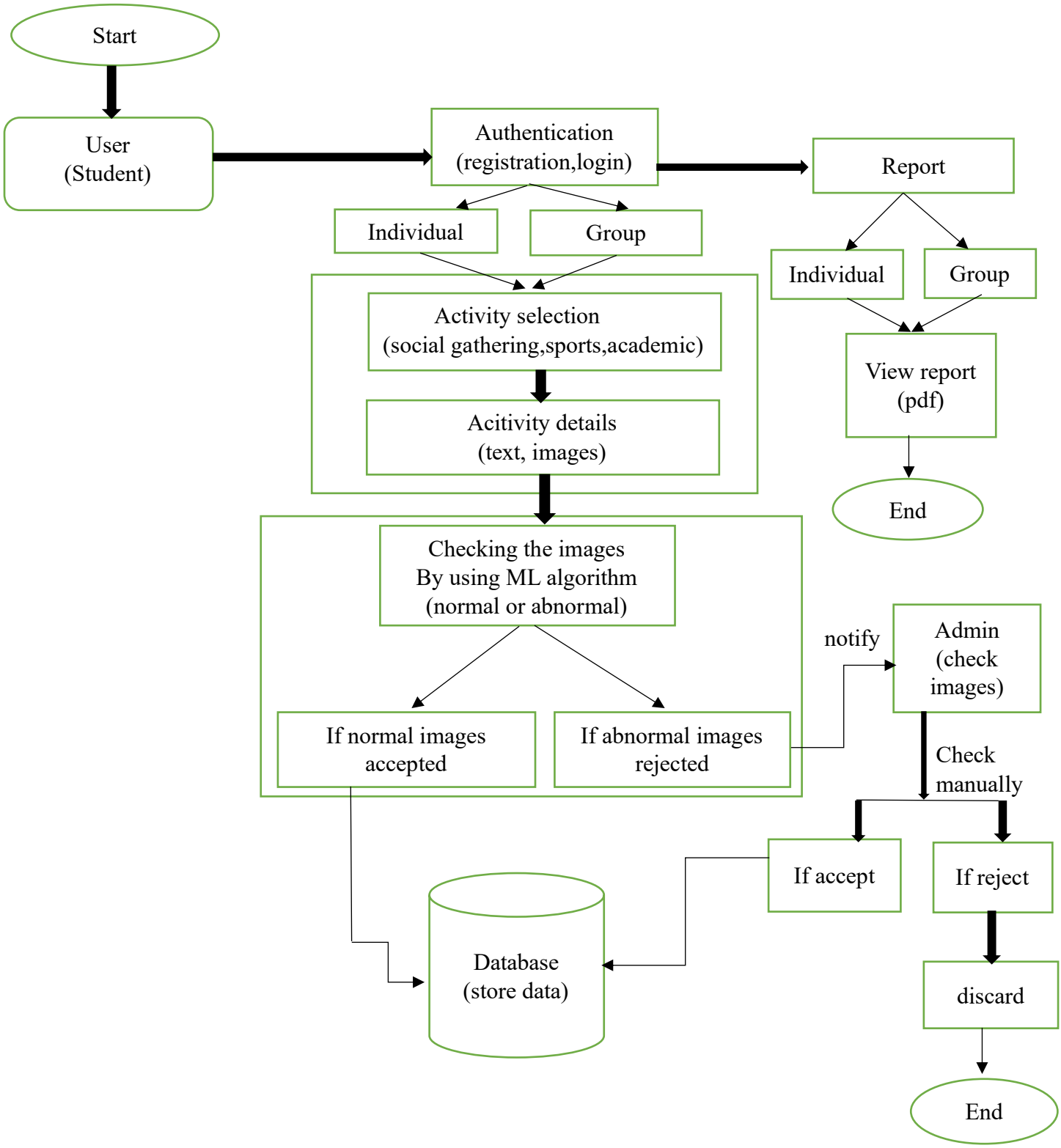
1. Deployment – Phased rollout.
2. Maintenance – Updates, bug fixes, support.

(10) **Project Overview :-**

EduReport is a smart web-based system designed to automate the annual report preparation process in educational institutions. It integrates data collection, ML-based content validation, secure storage, and structured report generation. By reducing manual effort and improving accuracy, the system ensures consistency, transparency, and efficiency across departments. The project contributes to digital transformation in academic administration and enhances decision-making through insightful reports.

(11) **Proposed Work :-**

(1) Block Diagram :



Description of the diagram:

1. User (Student)
A student starts by accessing the system.

2. Authentication (Registration/Login)
The student registers if new, or logs in if already registered.
Authentication ensures only authorized users can submit activities.

3. Activity Selection
The student selects the type of activity: social, sports, or academic.

4. Activity Details (Text and Images)
The student uploads descriptions and supporting images of the activity.

5. Image Verification using ML Algorithm
The system automatically checks uploaded images using a machine learning algorithm to detect whether they are normal or abnormal (valid or invalid content).

6. Decision Branch: Image Accepted or Rejected
If accepted: Data is sent directly to the database.
If rejected: The system notifies the Admin.

7. Manual Verification by Admin
The admin manually checks the rejected images.
If the admin approves, the data is added to the database.
If not approved, the image/data is discarded.

8. Database Storage
Approved data is stored securely in the database.

9. Report Generation
Students and faculty can view the report in PDF format.
Reports can be generated for individual students or groups.

(2) Modules :

1. User Management Module
Registration & Login
Handles new user sign-ups and secure logins.
Includes email verification or OTP (optional).
Authentication & Authorization
Role-based access (Student vs. Admin).

2. Activity Submission Module
Activity Selection
Students choose type: Social, Sports, Academic.
Data Upload
Input form for text description.
File/image upload (with size & format checks).
Client-side Validation
Ensures required fields are filled before submission.

3. Machine Learning Image Verification Module
Image Preprocessing
Resize, normalize, and clean images before ML model input.
ML Model Integration
Model to classify uploaded images as normal or abnormal.
Can be trained using CNN (e.g., TensorFlow/Keras or PyTorch).
Decision Engine
If accepted → send to DB.
If rejected → notify admin for manual review.

4. Admin Review Module
Notification System
Alerts admin about rejected submissions.
Manual Verification Dashboard
View rejected images, approve or discard.
Audit Logging
Keep records of all approvals/rejections for transparency.

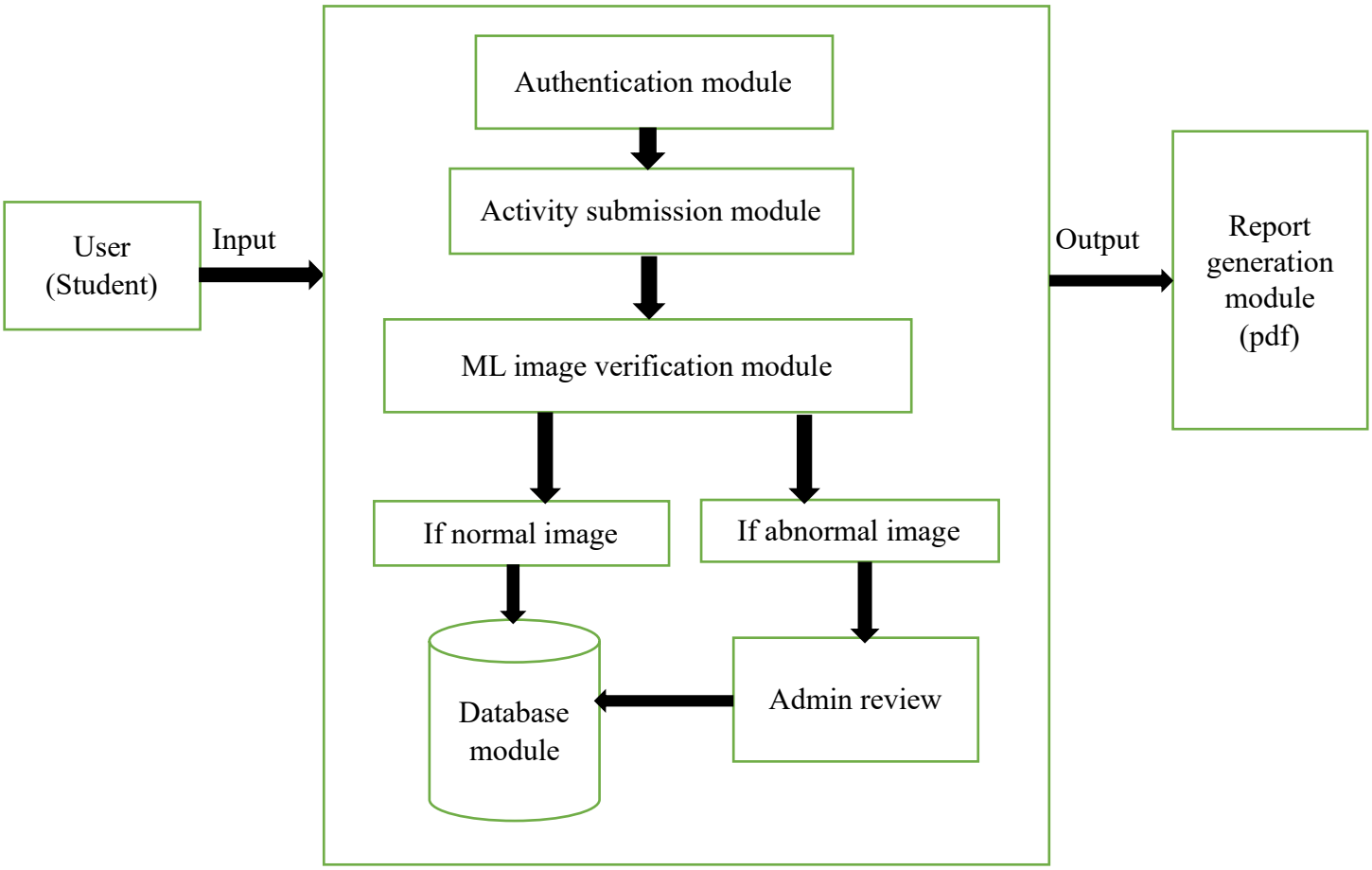
5. Database Module
Data Storage
User data, activity details, ML results.
Schema Design
Tables for Users, Activities, Images, Reports, Admin logs.
Database Management
MySQL / PostgreSQL / MongoDB depending on stack choice.

6. Report Generation Module
Report Type Selection
Individual report vs. Group report.
PDF Generation
Using libraries like ReportLab (Python) or jsPDF (JavaScript).
Download / View Interface
Secure access to reports for both users and admins.

7.Notification & Feedback Module

Email Alerts (SMTP, SendGrid, etc.)
In-App Notifications
Status Updates for students (Accepted, Rejected, Pending).

(3) Architecture Diagram:



Devlopment Environment :-

(3) Software Requirements :

Operating System:
Windows 10 or later / Ubuntu 20.04+ / macOS Monterey or later

Frontend Development:
Code Editor: VS Code /netbeans
Framework: React.js
Browser: Chrome / Firefox (for testing)

Backend Development:
Language: Python (Flask / Django) or Node.js

Package Managers: pip / npm

Database:
MongoDB (if NoSQL flexibility is required)

Machine learning:
Dataset: Kaggle
NLP: text processing

Reporting & Visualization:
Libraries: Chart.js, D3.js, Recharts

Version Control:
Git & GitHub (for code collaboration and tracking)

Other Tools:
XAMPP/WAMP (optional for local DB testing) Docker
(optional for containerization)
draw.io / Figma (for diagrams and UI design)

(4) Hardware Requirements :

Component	**Minimum Specification**
Processor (CPU)	Intel i3 / AMD Ryzen 3
RAM	4 GB 8 GB or more
Storage	256 GB HDD/SSD
Display	13" or larger Full HD Monitor
Internet	Stable broadband connection

(7) References :-

i.Journal/Conference Papers:
https://www.researchgate.net/publication/388826881_ANNUAL_REPORT_PORTAL_FOR_EDUCATI ONAL_INSTITUTES_Department_of_computer_science

ii.Web reference link:
https://www.ijaresm.com/annual-report-portal-for-educational-institutions

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